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Task Description:	NASA's Advanced Life Support program believes that particulate solid substrates are best suited to meet its short and long term needs for plant growth during space flight. Due to mass, volume and power constraints, root modules are characterized by small volumes of growth media and high root density. These conditions leave little margin for error in control of water, air and nutrition. Results from plant growth experiments utilizing particulate growth media during space flight have demonstrated the difficulties associated with providing reliable reproducible air and water (and thus nutrients) supply to plant roots. This is attributed to the general lack of understanding of fundamental transport processes occurring in the growth media. The quantitative characterization of physical processes associated with the flow of wetting and non-wetting phase(s) in particulate plant growth media in microgravity is essential for the successful design and control of plant production systems during space flight. It has recently become evident that the macroscopic dynamic behavior of the temporal and spatial under-saturation levels in particulate media strongly depends on the physical behavior of the fluid at the microscopic (pore-scale) level. Ultimately, the stability and break-up of liquid-bridges, -columns, and -films, dynamic rearrangement of particles, contact angle dynamics and hysteresis, phase discontinuities, capillary fingering and entrapment of the non-wetting phase together with the time-dependent complex geometry of porous media will determine the saturation levels and its stability on macroscopic (integral) scales. However, it is not yet obvious if and how much some of these micro-scale phenomena are gravity dependent.Therefore, we propose a comprehensive program of research that will employ some new advanced experimental, numerical and analytical techniques for examining flow and distribution of fluids in plant growth media. For example, Nuclear Magnetic Resonance Imaging (MRI) will provide a visual record of wetting front dynamics and gas diffusion in porous media. New mathematical models that account for adsorption and liquid films, discontinuous phases, effect of vibration, solitary ganglion, pore network behavior, fingering, capillary instabilities, air entrapment, medium anisotropy and heterogeneity, and the hyperbolic nature of moisture distribution will be evaluated. Upscaling procedures will be employed to describe microscale phenomenon on scales relevant to root modules. Hydraulic conductivity and water retention characteristics will be experimentally verified under simulated microgravity conditions using terrestrial zero-g simulators. The final conceptual model of water and air transport will be verified in 1 g by forward and inverse simulation on the root module scale using flight compatible measurement techniques. The research will result in a model and experimental methods that are ready for final validation in microgravity. Improve control and monitoring of water delivery to plants in microgravity Improve plant growth systems for microgravity crop production and research. Reduce or eliminate environmental control issues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will provide scientific knowledge of fundamental mechanisms and processes related to water and air movement through plant growth media in microgravity. In order to achieve this goal current earth-based conceptual models of flow through porous media will be critically analyzed. We expect this work will make significant contibutions to understanding flow phenomenon on earth. This information will be of interest to hydrologists, agricultural engineers, soil physicists, and agronomists/horticulturists who are interested in water and air transport through soil (porous media) or removal of soil contaminants.
	In 2002 this NASA funded multidisciplinary team began to address flow and distribution of fluid phases in porous plant growth media in microgravity. This research effort has identified several problems ranging from inadequate characterization of media properties under variable gravity, to phase entrapment and onset of flow behavior not representable by standard flow theories. Some of the key issues involve: Characterization of porous plant growth media--- Optimal set points for water delivery in any g environment strongly depend on growth-media hydraulic properties. Although baked ceramic aggregates have been used in space flight for over 10 years, the team conducted the first rigorous characterization of physical and hydraulic properties. This work has enhanced the understanding of aggregate dual porosity, pore size distribution, water conducting porosity, air entrapment, and air entry that allows critical examination of water delivery set points that have been used in space flight. Physical and hydraulic characteristics were established for the 0.25-1 and 1-2 mm baked ceramic aggregates commonly used for space flight. Wet and dry packing increased bulk density in comparison with a control. Dry packing resulted in highest densities; minimum and maximum densities (1 g) for dry packing were established. Magnetic resonance imaging (MRI), Mercury porosimetry, and water retention analysis was used to quantify total porosity, wettable porosity obtained with natural saturation (satiation), and pore size distribution of the macro and micro (intra aggregate) pore space. MRI was used to establish the REV for porosity, indicating the transition from micro scale to Darcy (scale at which Darcy's law is valid). Saturated and unsaturated conductivity of baked ceramic aggregate was measured for the root module scale using the earth based microgravity analogue of horizontal flow through a shallow depth of porous medium. Saturated and unsaturated hydraulic conductivity were directly measured by two methods using flight compatible methods and experimental design. Hydraulic conductivity predicted using the RETC program (USDA, Riverside, CA) and commonly used Van Genuchten/Mualem models compared favorably with direct measurements for inter aggregate pore space. The water retention characteristic of inter-aggregate (macropores) was quantified for wetting and drying porous medium; the water retention characteristic for drying porous medium was measured for the entire range of water contents. Additionally, stable reference water content was established for drainage of the macro pore space. Water retention remains a key physical property of interest with recent KC-135 measurements providing some of the very limited microgravity datasets to date. Issues related to finite volume and small spatial scale--- Space flight limitations on mass and volume result in root modules characterized by small volume of growth media and high root density. The small spatial scale accentuates the impact of inherently irregular wetting patterns. Wetting front irregularities may span the entire root module depth promoting entrapment and discontinuous phase formation. Root modules used for space flight contain a finite volume of plant growth medium. The effect of the physical presence of plant roots on bulk density, water retention, and hydraulic conductivity were evaluated for finite volumes of baked ceramic aggregate. Results indicate that penetration of plant roots into 1-2 mm baked ceramic aggregate creates a new porous medium with dynamic physical and hydraulic properties that change with increasing root volume. Consequently, water management based on macroscopic criteria (set points) may be constantly changing. Use of unrooted porous medium physical and hydraulic characteristics or sensor calibrations to interpret data from rooted porous medium may result in inaccurate measurement of porous medium water content. Micro- and reduced gravity effects on water transport--- The distribution of water and air in porous medium under microgravity was addressed using Hexadecane, a fluid immiscible with and lighter than water. Changes in pore size distribution resulting from vehicle vibration and/or particle separation due to microgravity was examined for baked

Task Progress:	ceramic aggregate. Aggregate containing residual Hexadecane saturation was subjected to vibration. The effect of vibration was to increase the likelihood breakup and mobilization of blobs entrapped in porous medium at residual saturation. Blob size distribution was adequately represented by the Van Genutchen expression. On-going modeling efforts are aimed at understanding how microgravity may affect discontinuous phase mobilization and distribution in porous medium. The effect of gravity on pore fluid distribution was also evaluated during KC135 flight on glass beads wetted to residual saturation with Hexadecane. Results indicate that separation of individual beads did not occur; however, groups of particles with encapsulated Hexadecane blobs did rearrange themselves during 0g. Movement of individual particles or clusters in microgravity will affect pore size distribution; the effect of this movement on discontinuous clusters of air or water is the subject of on-going investigation. Water moves within porous media under the influence of pressure, gravity and surface/capillary forces. Surface/capillary forces dominate in micro- or low gravity environments when both water and air phases are present. Measurements aimed at furthering the understanding and characterization of fluid flow and distribution during infiltration and steady flow were made at the micro and sample scale. Magnetic resonance imaging was used to obtained 2-D images of infiltration into horticultural foams and particulate medium. Video microscopy documented water/air organization and reorganization, infiltration and drying and particle separation during the variable gravity of KC135 flight. Wetting front irregularities or surface forces that cause water and air to segregate into wet and dry clusters are difficult to control or predict. Irregularities in pore size distribution and expansion of capillary length scales enhance phase entrapment in 0g. Once these clusters become established during initial infiltration they may be hard to remove in 0g due to the combination of water-primed pathways and dominance of capillary forces. Additionally infiltration into pre-wetted porous media occurs faster in 0g than in earth gravity. On-going modeling efforts continue to address infiltration patterns observed in 0g and 1g. Saturated hydraulic conductivity for baked ceramic aggregate and glass beads was confirmed to be the same in 1g, 2g, and 0g. In addition, mini tensiometers were developed and tested for use in quantifying the energy of water and matric potential change in response to variable g. Matric potential equilibration during 0g was faster in 1-2 mm than 0.25-1 mm aggregate, and was attributed to a higher macro pore hydraulic conductivity in the larger particle size aggregate. Time domain reflectometry (TDR) was also tested and used to measure porous medium water content during variable g. Combined use of these sensors allowed measurement of water retention characteristics during variable g of KC135 flight. Analysis of the data is ongoing, but preliminary results indicate that the matric potential for which air entry occurs is reduced in 0g than 1g. Modeling--- Modeling efforts have addressed hysteresis in the water retention characteristic by calculation of the capillary-pressure versus saturation curves for idealized spherical particle arrays. Capillary rise in non- spherical pores of various configurations was also predicted. Simulations of the effect of gravity on liquid vapor interfacial configurations has shown that gravity becomes significant only for pores >>1mm and potentials close to zero. The Lattice Boltzman modeling approach was used to address gaseous diffusion through porous medium in 0g and 1g. Results have shown that interpretation of 1g simulations requires analysis of data from both a horizontal and vertical perspective. Simulated gas diffusion was only slightly lower in 0g than 1g. Development of a model to simulate multizone, multicontinuum, multiphase and multispecies transport is on-going. The simulator uses MathLab as the platform and is designed to easily incorporate a wide variety of processes by adding “plug and play” modules. The simulator is grounded in principles of compositional multiphase flow and is intended to be an extensible platform for considering environmental processes. The simulator package has three components: a graphical user interface (GUI), a numerical engine, and an extensible set of self-contained modules that communicate with both the GUI and the numerical engine. Governing equations consider a wide range of processes in order to guide strategy and code development. These include, but are not limited to, transport of water and gas in a vegetated medium, gas species in a mixture, atmosphere adjacent to porous medium and plant dynamics. Current emphases is on improving the GUI, the numerical engine and a selected set of modules that will allow consideration of non-isothermal water and gas flow in a porous medium. Water content sensing and control--- Matric potential or water content sensors have been used to control and schedule water delivery to plant growth media. The number of flight compatible sensors currently limit measurement of porous medium hydraulic characteristics in microgravity. Evaluation of the measurement volume (what the sensor “sees”), accuracy and repeatability of the dual probe heat capacitance sensor is on going. This sensor does not rely on empirical calibration, but rather computes porous medium water content from the physical properties: density and heat capacitance of porous medium solids and water. It also has a lower sensitivity to changes in porous medium bulk density. Limited sensing volume of most sensors used in root modules (<1 cm3) combined with discontinuous phases (described above) may result in “accurate” readings on the mesoscale (several pores or particles), but an inaccurate assessment of water content in the bulk volume of growth media. Summary --- The effect of gravity on fluid flow through porous medium is most significant at the cluster scale (cluster of several particles “glued” together by water). Cluster scale instabilities may become apparent in 0g but not 1g, such as water entrapment in larger pores in 0g. Increased phase entrapment results in increased hysteresis. In microgravity the capillary length scale increases supporting capillary flow in larger pores as compared to 1g. Increased length scales may also lead to fingering during infiltration or drainage. Use of medium with broader pore size distributions may reduce the cluster scale affects caused by 0g. At the root module scale the most significant effect of 0g is the lack of hydrostatic gradient of water and air. Local (cluster scale) configurations of air and water will be controlled by capillary forces. Investigation of how changes in air and water distribution at the cluster scale translate into averaged hydraulic parameters at the root module scale is on going. The Richards equation continues to provide a good description of average flow characteristics on the root module scale. Interpretation of existing data from microgravity is currently limited by inadequate characterization of the root module system in earth gravity. Most significant has been lack of consideration of the impact of root volume in a finite volume of porous medium and understanding output from flight compatible water content or matric potential sensors.
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